

Mohit Kumar Sharma

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Summary

Software Engineer with strong expertise in building scalable applications and deep interest in leveraging AI and emerging technologies. Driven by a growth mindset, I continuously upskill and contribute to innovative and high-impact solutions.

Education

Indian Institute of Technology (BHU) Varanasi

B.Tech - Electronics Engineering

2020 - 2024

CGPA - 8.42

Experience

Software Engineer

Microsoft

Hyderabad, India

July 2024 – Present

- Designed and implemented a stale-while-revalidate caching layer (in-memory + on-disk + asynchronous background revalidation) for an expensive video encoder capability query in a real-time communications media stack, **eliminating a 350ms** blocking call on application startup.
- Drove H.264 decoder hardware utilization on an Intel GPU Model from 20% to 70%+, lifting overall Microsoft Teams video hardware utilization from **88% to 90%**, by pinpointing root causes through field-telemetry analysis and partnering directly with Intel to whitelist qualified drivers.
- Designed and built a scalable distributed test execution and analytics system with real-time test tracking, automated failure analysis, intelligent alerting, and centralized dashboards, significantly reducing manual debugging effort and improving large-scale issue detection and overall test infrastructure reliability.
- Introduced new telemetry metrics tracking hardware failures and encoder/decoder frame drops, enabling continuous monitoring and faster regression detection.
- Led validation of a terminal-based application used by hardware partners to certify GPU drivers against Microsoft Teams video calling across H.264, H.265, and AV1 codecs. Delivered two major releases and stabilized 1,600+ test cases, eliminating crashes/failures and establishing a critical release gate safeguarding call quality for millions of Teams users worldwide.
- Exposure:** C++, Python, Multithreading & Concurrency, System Design, KQL, Power BI

Software Engineer Intern

Microsoft

Hyderabad, India

May 2023 – July 2023

- Designed and developed multi-turn conversational interactions for the TAC Co-pilot, enhancing UI/UX by implementing intuitive chiclet-based input options and aligning the interface with the M365 Co-pilot design system.
- Leveraged the Microsoft LLM framework to accurately interpret user queries by identifying intents and extracting entities, improving overall conversational performance and user experience.
- Exposure:** React, TypeScript, REST APIs, UI/UX, Prompt Engineering

Full-Stack Developer Intern

Warelogg

Remote

Dec 2022 – Dec 2022

- Enhanced the Warelogg React web application by improving UI/UX, responsiveness, and overall functionality to deliver a smoother user experience.
- Implemented Node.js-based backend updates to manage and modify warehouse details, ensuring secure and accurate data handling through admin-level access controls.

Projects

Picture Manager Application [🔗](#)

- Developed a web-based application featuring a secure login system powered by Face Recognition technology, integrating JWT-based authentication and authorization to safeguard user/admin access and protect all API endpoints.
- Developed intelligent photo-management functionality using facial recognition to automatically categorize images into individual and group folders, and implemented event-related CRUD operations with MongoDB for optimized data handling.
- Exposure:** Next.js, Node.js (Express.js), Face-api.js, MongoDB, Tailwind CSS

Technologies

Languages: C/C++, Python, JavaScript, React, NodeJS, SQL, NoSQL

Core Skills: Full-Stack Development, Data Structures and Algorithms, Problem-Solving, OOPS, DBMS, Agile Methodology

Achievements

- Solved 800+ coding challenges on platforms like LeetCode, GeeksforGeeks, showcasing strong problem-solving skills.